



PREVALENCE OF ASTHMA DISEASE: IN CASE OF OBORRA PRIMARY
HOSPITAL OF OBORRA TOWN, HOROGUDURU WOLLEGA ZONE,
ETHIOPIA.

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CHAPTER ONE

1. INTRODUCTION

1.1. Background of the study

Asthma is chronic inflammatory disease of respiratory system (Global Initiative for asthma,2009).The chronic inflammation is associated with air way hyper -pensitivity or an exaggerated air way narrowing response to, such as allergens and exercise, that lead to recurrent symptom such as wheezing, dispend (shortness of breath),chest tightness and coughing. Several factors have been proposed to account of the cause of asthma disease and increase the prevalence (Jorge and Mazza, 2011).It is agreed that genetic changes in population would be too slow to account. Environmental factors have been a key role aogenesis of asthma disease. Among the environmental determinant exposure to allergens such as house dust mite, grass pollen, cockroach and animal dander (Bourdon *teal*, 2009).

Various methodologies have been used by researchers to measure adherence. These include the Medication Event Monitoring System (MEMS), patient self-report, patient pill-counts, pharmacy databases or refill rates, and blood level assessments. The MEMS uses a microprocessor technology that has been embedded in the cap of a container to record time and date the container with medication was opened (Aslani & Krass, 2009). Using the MEMS can generate incorrect results from patients who may remove more than the single recommended dose or may not remove a dose at all when they opened the bottle. (Aslani & Krass, 2009; Carter *et al*, 2010).

The global status report on non-communicable diseases (NCDs) indicates that NCDs kill more people annually than all other causes combined (WHO- GSR-NCDs, 2010). It noted that on the contrary to popular opinion 80% of NCD deaths occur in the low and middleincome countries and that much of the human and social impact caused annually by NCDrelated deaths could be averted through well-understood, cost-effective and feasible interventions (WHO- GSR-NCDs, 2010)

Investigator recently reported that in Canada, five thousand people die each year by this asthma disease (Kim and Mazza, 2011) by .And also asthma account for more absence from the school and more hospitalization of children and reproductive man power at risk by this asthma disease. Moreover, in developing country like Ethiopia, the effect of this disease is more serious

(Yemaneberhan *et al*, 1997). Particularly, Oborra town is palace where several peoples are affected by asthma disease. Therefore; this study was assessing the factor that contributes for the cause of this disease. Asthma disparities have been attributed to differences in social economic status, but in African community when compared to Caucasian communities due to variety of social factors including disparities in health. More recent research has shown that other factors are more important in this inequality. There is a higher prevalence of poorer health outcomes of patient's asthma case assessments and communications, the social structure prescribed to patient (Bryanth Stephens, 2009). Additionally, studies by rickets have shown that doctor (providers) prescribed the appropriate amount of medicine, at least the third of the patient would not take their medicines due to cultural and health belief (Bryant Stephens, 2009).

Haynes and colleagues (2008) reported that patients who are prescribed self-administered medications usually take less than half the prescribed doses (Haynes *et al*, 2008). Sub-optimal adherence to inhaled medicines for COPD and asthmatic conditions are reported by Mika colleagues (2013), to be associated with poor symptom control and reductions in health-related quality of life, in addition to a higher healthcare utilization and cost (Mika *et al*, 2013)

Another reason why African community have a higher rates of adults asthma is because residential segregation. One article illustrated those poor African communities that are separated from other ethnic communities has a higher concentration of poverty which leads to a lack of resource and support. According to in western Ethiopia asthma accounted for 2% of outpatient and 5.4% of medicinal admission (Seyuom B, 1992).

As Ethiopia in General, Oborra town particularly, is palace where several peoples are affected by asthma disease. Therefore; this study was assessing the factor that contributes for the cause of this disease. Asthma disparities have been attributed to differences in social economic status, but in African community when compared to Caucasian communities due to variety of social factors including disparities in health. More recent research has shown that other factors are more important in this inequality.

1.2. Statement of the problem

Asthma is the most common respiratory disease in developing countries and developed countries. Asthma is a very difficult disease that associated with other chronic respiratory disease

(Kim and Mazza, *al.*, 2011) and the symptoms of the asthma also very similar with other respiratory disorder disease like TB, Common cold and etc. Several factors have been proposed to account for the cause and its prevalence, even though environmental factors play a key role. Furthermore, asthma disease can affect all aged group of people all over the world (Kim and Mazza, 2011). Therefore, this study was attempted to answer the following questions: regarding to this study

1.3. Research questions

- ✓ What are the factors that contribute for the cause and effect of asthma disease?
- ✓ How the symptoms of asthma disease are identified from other respiratory disease?
- ✓ What is the main factor for the prevalence of asthma disease?

1.3. Objectives of the study

1.3.1 General objective the study

To assess the health impact of asthma disease in Oborra hospital.

1.3.2 Specific objectives

1. To determine the prevalence and effect of asthma disease in Oborra town.
2. To identify the symptoms of asthma disease are identified from other respiratory disease
3. To identify which environmental factors are the main cause for asthma disease.

1.4. Significance of the study

Asthma is the worldwide problem affecting more than 230 million people around the globe. Therefore the finding of this study will provide enough information about the factors that aggravate asthma disease, Give awareness about the symptoms and effect of asthma disease, Be the base line for the further related study

CHAPTER TWO

RELATED LITERATURE REVIEW

2.1. Concepts of Asthma Disease

Asthma is a chronic disease characterized by recurrent attacks of breathlessness and wheezing (World Health, 2000). Asthma remains the most common chronic respiratory disease (Ottawa and Ontario, 2007) affecting approximately 10% of the population. Although, asthma is often believed to be a disorder localized to the lungs, current evidence indicates that it may represent a component of a systemic airway disease involving the entire respiratory tract, and this is supported by the fact that asthma frequently coexists with allergic rhinitis. Poor asthma control contributes to unnecessary morbidity, limitation to daily activities and impairment in overall quality of life (Bourdon *et al.*, 2009). The chronic inflammation associated with airway hyperresponsiveness (an exaggerated airway narrowing response to triggers such as allergens and exercise), dyspnea (shortness of the breath), chest tightness and coughing.

A Swedish study also found increased prevalence of physician diagnosed asthma and an unchanged prevalence of most respiratory symptoms over an 11 year period that concluded that social awareness and changes in physician diagnostic practices may be the possible causes of increasing asthma reporting (Ekerljung *et al.*, 2011).

In a systematic review of empirical studies in adults published between 1999 and 2009 conducted in France by Charles and colleagues (2011), to investigate the associations between perceptions of asthma and treatment beliefs and also adherence to inhaled corticosteroids. They noted that better adherence rates were associated with perception of the chronicity of asthma and its consequences on daily life, as well as the concept that it is necessary to continue treatment in the absence of symptoms. On the contrary, poor adherence rates were associated with the fear of side effects and the belief that treatment is ineffective in controlling symptoms (Charles *et al.*, 2011).

Symptoms episodes are generally associated with appropriate treatment (Fitz Gerald *et al.*, 2012). Asthma is associated with T-helper cell type 2 (Th2) immune response (Kim and Mazza, 2011) are typically of other atopic conditions various allergic such as dust mite, cockroach residue, furred animals, molds, pollen and non-allergic such as infections, tobacco smoke, cold air, exercise triggers produces a cascade of immune mediated events leading to chronic airway inflammations. Inflammation mediators such as histamine and cysteinyl leukotrienes that cause

bronchospasm (contraction of the smooth muscle in the air way), edema (swelling) and increase mucus secretion (Fitz Gerald *et al.*, 2012).

The mediators and cytokines released during an early phase of an immune response to an indicating allergen triggers a further inflammatory response(late phase asthmatic response)that lead to further air way inflammation and bronchial hyper reactivates(Lemansket *al.*,2011).The diagnosis of asthma should be suspected in patient with recurrent cough, wheeze, chest tightness and shortness of the breath such variable symptoms occur upon exposure to allergen or irritant, that worsen at night and that respond to appropriate asthma therapy are strongly suggestive of asthma Kaplan (*eta.*,2009). Alternative cause susceptible asthma symptoms should be excluded such as chronic obstructive pulmonary disease, bronchitis, chronic sinusitis gastro esophageal reflux disease, recurrent respiratory infection and heart disease.

A positive family history of asthma or other atopic disease and/or personal history of atopic disorder particularly allergic rhinitis, can also be helpful in identifying patient with asthma (Few *et al.*, 2010), Exposure to agents encountered in the work environment can also cause asthma. The diagnosis of asthma in young children is often more difficult since episodic wheezing and cough are common and spirometry is a reliable in patients under 6years of age.

A useful method of confirming a diagnosis in young children is a trial treatment with short acting bronchodilators (Kim and Mazza, 2011). Spirometry is the preferred objective measure to assess for the reversible respiratory system obstruction (Yemanehberhan *et al.*, 1997),(i.e. rapid improvement in lung function after elation of rapid acting bronchodilators and confirm a diagnosis of asthma).

During spirometer the patient is instructed to take the deepest breath possible and then to exhale as hard as fully as possible into the mouth piece the spirometry.Avoidance of relevant allergens or irritants is an important components of asthma management(Mengistu *al.*,2011) patients allergic to house dust mite should be instructed to use allergen impermeable covers for bedding and keep the relative humidity in the home below 50%(to inhibit mite growth).According to the recent estimates, asthma affects 300million people(Peter Costa,2007).Although people of all age suffer from the disease ,it most often starts childhood, especially children who have low birth weight,who are exposed to tobacco smoke and are raised in a low income environment. According to the latest data published asthma death in Ethiopia reached 6,743 or 1.12% of total death (World Health Organization, 2014).

3. RESEARCH METHODOLOGY

3.1. Description of the Study area

The study will be conducted in Oromia regional state, Horo Guduru Wollega, particularly Oborra town primary Hospital, Eastern Ethiopia. Oborra town is located at 400km away from Addis Ababa, which is the capital city of Ethiopia. All the soils types have good agricultural potentialities; annual rain fall varies between 1300mm and 1700mm respectively (CSA, 2013).

3.2. Study Design

The study will be conducted on the patient's document of 2015 E.C in Oborra primary Hospital to determine the prevalence of asthma allergic disease among patients. Survey method of study design will be used.

3.3 Source of Data

The source data will use secondary data which recorded in cards of patients who were visited at Gibe Hospital.

3.4. Study Population

Study population of research include all asthma diseases patients who will treated in oborra town and who going to visit in oborra primary hospital and Oborra town peoples who will assessable during the data collection.

3.5 Sample size and sampling technique

In this study the sample size will be include all asthma diseases that register in log book patient recorded. Random sampling technique will be appropriate to get the data; particularly purposive sampling technique will be used on target individual to get information about typhoid fever disease.

The sampling technique and sample size of the study will be determined by using the following formula. Where, z = standard normal deviate at 95% level of confidence that is 1.96, $q=1-p=0.05$, d is margin of error 0.1. Therefore, the sample size calculated for residents is 74. Among the total population of oborra town 74 of them will be selected as a sample. To distribute the questionnaires the person who come first was serve first

To distribute the questionnaires random sampling techniques and the sample for study (Martin, 1995). Sample size determination formula was be used.

Where, n =sample

d =margin error

z =confidence level is equal to 1.95

p =population proportion

$q=1-p=0.05\%$

$$n = \frac{Nz^2pq}{d^2(N-1) + z^2 + pq}$$

$$n = \frac{2434(1.96)^2 + (0.05)/(0.05)^2(2434-1) + (1.96)^2(0.95)(0.05)}{74}$$

$$n = 74$$

3.6. Data collection tools and methods

This research will be conducted by using both primary and secondary sources of data. The primary data will be collected using questionnaires which were closed ended questions and interviews. The secondary data will collected by introducing the objective of the study to obborra hospital. Then experts in the health institution will asked for giving a patients registration card and depending on the registered information of the year from 2022-2023, the patient s data will be taken and the status level will be known.

3.7. Data Analysis

The collected data will be analyzed using data and other relevant statically tool. The results will be present in the form of tables, graphs and pie chart. The data will analyzed by using tables and percent.

4. Work plan and budget

4.1. Time schedule

Started schedules	April	may	June	July
Activities	✓			
Title selection		✓		
Introduction of the study		✓		
Literature review		✓		
Methodology of the study		✓		
Data collection			✓	
Proposal submission				✓

4.2. Budget Break Down

Number	Instrument	Measurement unit	Cost in terms of birr	Aggregate cost
1.	Pen	1	5	5
2.	Internet	24hr	.50	400
3.	Paper	20	0.25	5
4.	Print	18	1	18
5.	Flash	1	165	428

5. Reference

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Dear Respondents

This questionnaire is designed to help a study an Assessment of the Health Impact of Asthma disease (in case of primary Hospital of oborra town), n. This information is only used for academic purpose. So, I request you to answer all the questions below to the best of your information, in which please tick your choice in the box to show your response, and write down your own opinion, reason and suggestions in the space provided. Your information will be handled with the highest order of confidentiality. Please don't write your name anywhere on the questions. I thank you for spending of your time to complete this research questionnaire.

I. General Information

1. Age -----

2. Work office-----

3. Sex----- A. Male B. Female

4. Educational Status

A. illiterate

B. 9-12Grade complete

C. 1-4 Grade Complete

D. Diploma

E. 5-8 Grade Complete

F. Degree and above

5. Work experience -----

6. Department/ section-----

Questionnaire: - for assessment on the factors that contribute for the cause of asthma disease in oborra town.

Part one: - personal information

Age_____

Job: A. Student ☐ B. Employer ☐ C .Merchant ☐ D .Driver☐
E, another ☐

Religious: A. Muslim ☐ B.Orthodox ☐ C. Catholic☐ D. Protestan☐ E.
Other ☐

Ethnicity: A. Oromo ☐ B. Amhara ☐ C. Garage ☐ D. other ☐

Family status: A. Single ☐ B. Married ☐

Part two:-

1 Do you have any respiratory disorder disease?

A. yes ☐ B. NO ☐

2 If your answer for question number 1 is yes, what is it?

A .Common cold ☐ B. Allergy ☐ C. Asthma ☐ D. other. ☐

3. If your answer for question no 2 asthma, what do you think the factor that causes asthma?

A. tobacco smoke ☐ B. Environmental pollution ☐ C. Dust mite ☐
D. cold air ☐ E. exercise. ☐

4. At what age you have suffered to this asthma?

A. at child age ☐ Bat adult ☐ C. old age ☐

5. When it is difficult to you?

A .at night ☐ B, in the morning ☐ C.during the rain ☐ D. during warm
cctemperature. ☐

6. What are the symptoms of asthma disease?

7What your Environmental hygiene is look like?

A. clean ☐ B polluted ☐ C. somewhat clean ☐

8. If your answer for question no 7 is B what makes it or what is the reason behind?

A. Gass from burned waste ☐
mals in your house?

A. YES ☐ B. NO ☐

10. How do you remove the domestic solid waste from the environment?

A. By burning ☐ B. by simply discarding at far away from the environment ☐

C. By accumulating and using as fertilizers latter ☐

11. How do you remove the domestic liquid waste?

A. By piping to the river ☐ B. by accumulating in the whole ☐ C. By treating ☐
and recycling ☐

12. Which environment do you think is the main factor to cause asthma?

A. Net and clean ☐ B. Polluted by different unpleasant smell ☐ C. environment ☐
which have polluted by waste from the factor ☐

13. How the environment could be neat and clean?

14. Have you ever contact a doctor?

A. yes ☐ B. no ☐

15. If yes, what your doctor recommend you?

16. What kind of medication do you use to control your asthma?

A. oral tablet ☐ B.in haler oxygen ☐ C. liquid by syringe ☐ D. not
exercising ☐

17.What are the impacts of asthma in your life beside its health impact?

A.Social impact ☐ B. Economic impact ☐ C. Psychological impact ☐